1. Which tense is used to state universal scientific facts and general laws?

2. In 'DNA replication occurs in the 5' to 3' direction', the tense is:

3. Which tense is primarily used in the Materials and Methods section?

4. In 'The cells were incubated at 37 deg C for 24 hours', the tense is:
Passive)

5. Which tense connects past research discoveries to the current state of knowledge?

6. Identify the present perfect tense in the following sentences:

7. A Zero Conditional sentence expresses:

8. In 'If water boils, it changes into vapor', which conditional type is this?

9. A First Conditional sentence typically expresses:

10. Choose the correct verb: 'If the enzyme is heated above 80 deg C, it ___.'

11. Choose the correct verb: 'If you add ethanol, the nucleic acid ___.'

12. Which conditional type uses past simple in the 'if' clause and 'would + verb' in the main clause?

13. In 'If we used a different vector, the yield might improve', the type is:

14. A Third Conditional sentence expresses:

15. Choose the correct verb: 'If the temperature ___ lower, the crystals would not have degraded.'

16. In 'Figure 2 shows the protein binding kinetics', 'shows' is in:

17. Which tense is formed by 'had + past participle'?

18. In 'Before the assay ended, the control had already cleared', the tense is:

19. Identify the past continuous tense:

20. Which verb form represents the present perfect continuous tense?

21. Stative verbs (e.g., consist, contain, belong) generally do NOT take:
Progressive) Aspect

22. Which of the following sentences violates stative verb rules?

23. In formal subjunctive conditional clauses, what verb replaces 'was' for all persons?

24. Select the correct subjunctive form: 'If the inhibitor ___ present, the reaction would cease.'

25. Which auxiliary verb indicates future intention or projection in scientific research?

26. In 'By next year, the lab ___ genomic sequencing on 500 samples', the verb is:

27. What tense is 'will have completed'?

28. Identify the sentence written in the simple future tense:

29. In 'The scientist said that the bacterium ___ Gram-negative', why is present tense retained?

30. Choose the correct conditional form: 'If the primer ___ to the template, extension occurs.'

31. Identify the tense: 'Researchers have been investigating CRISPR mechanisms since 2012.'

32. Which conjunction can introduce a negative conditional clause meaning 'if not'?

33. Complete the sentence: 'Unless the pH is maintained at 7.4, the cells ___.'

34. In 'Had we known the sequence, we would have ordered primers', what is omitted?

35. The construction 'Had we known...' is an example of:

36. Choose the correct verb: 'The technician ___ the autoclave before the power failure occurred.'

37. Which tense is used in reporting specific results obtained in a completed trial?

38. In 'Table 3 summarizes the pharmacokinetic parameters', the tense is:

39. Choose the correct verb: 'When glucose is exhausted, yeast ___ to ethanol fermentation.'

40. Identify the incorrect verb form in scientific conditional sentences:

41. Select the sentence with the correct tense for literature review in an Introduction:
1953) has synthesized amino acids.   (B) Miller (1953) synthesized amino acid
1953) is synthesizing amino acids.   (D) Miller (1953) will synthesize amino aci

42. Select the correct tense when referencing general scientific consensus without a specific date:

43. Identify the correct mixed conditional sentence:

44. Choose the correct verb: 'By the time the incubation completed, the OD600 ___ 1.5.'

45. Identify the erroneous tense shift in: 'We added the antibody, vortexed the tube, and then we measure absorbance.'

46. Choose the correct inverted conditional form for: 'If the pressure should rise beyond 2 bar...'

47. Choose the correct verb: 'If the buffer ___ prepared correctly, the protein would not precipitate now.'

48. Select the sentence showing correct backshifting in reported speech:

49. In 'Pasteur proved that spontaneous generation ___ impossible', the verb is:

50. Choose the correct verb: 'We observed that cell proliferation ___ exponentially during phase two.'

51. Which sentence demonstrates the future perfect tense correctly in experimental planning?

52. Identify the stative verb that cannot be used in the continuous aspect:

53. Select the sentence with correct conditional concord:

54. Choose the correct verb: 'The team ___ on developing monoclonal antibodies for over three years.'

55. In 'Were the recombinant protein toxic to the host, cell growth ___ cease':

56. Identify the sentence that correctly uses the past perfect to establish relative chronology:

57. Choose the correct verb: 'Provided that the temperature ___ constant, the enzyme follows Michaelis-Menten kinetics.'

58. Select the sentence with an ungrammatical conditional construction:

59. Why is 'If the cells will reach...' incorrect in standard English syntax?

60. Choose the correct verb: 'The bio-reactor ___ continuously since Monday morning.'

61. Identify the sentence with correct sequence of tenses in a complex sentence:

62. In 'Should you require additional viral vectors, please ___ the lab manager':

63. Choose the correct verb: 'If the reagent ___ contaminated, the spectrophotometer readings would have been erratic.'

64. Select the correct verb: 'Recent investigations ___ light on the mechanism of ribosomal stalling.'

65. In 'The graph shows that as substrate concentration increases, reaction rate ___':

66. Choose the correct conditional form: 'Had the centrifuge balanced properly, the rotor ___ not damaged.'

67. Identify the sentence that uses the present continuous tense legitimately for an ongoing experiment:

68. Choose the correct verb: 'Prior to our 2024 trial, no research group ___ synthesized this cyclic peptide.'

69. Select the sentence with an appropriate tense for presenting a standard mathematical equation:

70. Choose the correct verb: 'If the membrane ___ permeable, glucose would diffuse freely across the barrier.'

71. In 'As long as the pH ___ between 6.8 and 7.2, the enzyme remains active':

72. Identify the incorrect use of the present perfect in historical reporting:

73. Choose the correct verb: 'By the time the conference convenes, our paper ___ in Nature Biotechnology.'

74. Select the sentence where 'wish' is followed by the correct unreal past subjunctive:

75. Choose the correct verb: 'Unless the cells ___ protected by glycerol, freezing will cause ice crystal formation.'

76. Identify the sentence with an inappropriate tense shift between Methodology and Discussion:

77. Select the sentence with a flawlessly constructed inverted counterfactual conditional:

78. Choose the correct verb in an unreal conditional with inverted syntax: '___ the promoter active, GFP expression would occur.'

79. Select the sentence that correctly distinguishes between general empirical fact and historical observation:

80. Identify the sentence that violates the sequence of tenses in reported technical claims:

81. Choose the correct verb: 'It is essential that the sterile hood ___ decontaminated before retroviral handling.'

82. The use of the base verb 'be' in 'It is essential that the hood be decontaminated' is an example of:

83. Choose the correct verb: 'The committee recommended that Dr. Watson ___ the clinical trial director.'

84. Select the sentence with a grammatically correct mixed conditional representing past cause and ongoing present state:

85. Identify the subtle tense inconsistency in this experimental abstract sentence:

86. In 'No sooner had we initiated the reaction than the solution ___ turbid':

87. Choose the correct verb: 'If the enzyme were to aggregate, its catalytic efficiency ___ plummet.'

88. Identify the sentence containing an erroneous future auxiliary inside a temporal adverbial clause:

89. Why is 'As soon as the centrifuge will stop...' ungrammatical?

90. Select the sentence with correct sequence of tenses regarding a superseded prior hypothesis:

91. Choose the correct verb: 'The bioreactor ___ under aerobic conditions until the oxygen sensor malfunctioned.'

92. Identify the sentence using the past subjunctive correctly in a technical hypothetical statement:

93. Select the sentence that avoids ambiguity in stating ongoing literature findings:

94. Choose the correct verb: 'By the conclusion of this fiscal quarter, the bio-facility ___ 10,000 liters of recombinant insulin.'

95. Identify the sentence with an ungrammatical conditional inversion:

96. Choose the correct verb: 'The principal investigator insisted that the protocol ___ followed without deviation.'

97. In 'Scarcely had the culture reached log phase when the nutrient feed ___ exhausted':

98. Select the sentence with correct tense usage when presenting an illustration in an academic paper:

99. Choose the correct verb: 'If the cDNA library ___ contaminated, sequencing results would have shown chimeric reads.'

100. Identify the sentence that achieves flawless tense harmony, subjunctive mood, and academic scientific precision:

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Analyze the twelve-tense-aspect matrix of the English verbal system and evaluate its communicative precision in scientific discourse.

Q2. Detail the conventional tense shifts across the IMRaD sections (Introduction, Methods, Results, Discussion) of research papers.

Q3. Examine the grammatical rules and scientific applications of the Zero Conditional system.

Q4. Analyze the First Conditional system, focusing on real future conditions, probability, and modal substitutions.

Q5. Examine the Second Conditional system, highlighting hypothetical present/future states and the subjunctive 'were'.

Q6. Analyze the Third Conditional system and evaluate its utility in troubleshooting experimental failures and retrospective error analysis.

Q7. Discuss Mixed Conditionals and explain their syntactic formulation in correlating past causes with ongoing present realities.

Q8. Detail the syntax of inverted conditionals without 'if' (Had, Were, Should) and their stylistic prestige in academic publications.

Q9. Examine the Present Perfect versus Simple Past distinction in scientific citations and literature reviews.

Q10. Analyze the Past Perfect tense as an essential tool for establishing relative chronological priority in laboratory procedures.

Q11. Discuss the constraints on continuous (progressive) aspect with stative verbs in biochemistry literature.

Q12. Examine the Mandative Subjunctive in formal institutional, clinical, and laboratory safety protocols.

Q13. Analyze the sequence of tenses in reported speech and explain the 'Universal Truth' exception in biology papers.

Q14. Examine the Future-in-the-Past construction ('would' / 'was going to') in documenting historical research trajectories.

Q15. Analyze the Future Perfect and Future Continuous tenses in grant proposals and translational timeline projections.

Q16. Detail the syntax of 'Unless', 'Provided that', 'As long as', and 'In case' in biotechnology SOPs.

Q17. Examine common tense errors in the Results and Discussion sections of graduate biotechnology theses.

Q18. Analyze conditional inversion with 'Should' in laboratory emergency protocols and contingency SOPs.

Q19. Discuss the syntax and semantic difference between 'If I was' and 'If I were' in academic discourse.

Q20. Examine the Present Continuous tense: valid applications in live monitoring vs. common stylistic errors.

Q21. Analyze tense harmony in complex sentences containing adverbial time clauses ('after', 'before', 'until', 'as soon as').

Q22. Examine the syntax of 'wish' and 'if only' constructions in speculative scientific literature.

Q23. Analyze the sequence of tenses in defining equations, biological models, and software algorithms.

Q24. Detail the rules governing conditional sentences expressing concession ('even if', 'whether...or').

Q25. Analyze the past subjunctive in hypothetical comparison clauses introduced by 'as if' and 'as though'.

Q26. Examine the formation and usage of the Future Continuous tense in laboratory process automation.

Q27. Discuss subtle tense shifts when reporting historical controversies and scientific paradigm shifts.

Q28. Analyze the syntax and mechanics of conditional sentences expressing habit or typical routine behavior.

Q29. Examine errors in conditional sentence logic: the illicit insertion of 'would' in 'if'-clauses.

Q30. Analyze how choice of tense controls authorial voice, distance, and epistemic stance in scientific peer reviews.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [B]
Q3: [B]
Q4: [B]
Q5: [B]
Q6: [B]
Q7: [B]
Q8: [A]
Q9: [B]
Q10: [A]
Q11: [A]
Q12: [C]
Q13: [C]
Q14: [A]
Q15: [C]
Q16: [A]
Q17: [B]
Q18: [B]
Q19: [B]
Q20: [B]
Q21: [B]
Q22: [B]
Q23: [C]
Q24: [B]
Q25: [A]
Q26: [B]
Q27: [B]
Q28: [A]
Q29: [B]
Q30: [A]
Q31: [A]
Q32: [B]
Q33: [A]
Q34: [B]
Q35: [A]
Q36: [B]
Q37: [B]
Q38: [A]
Q39: [A]
Q40: [B]
Q41: [B]
Q42: [B]
Q43: [A]
Q44: [C]
Q45: [C]
Q46: [A]
Q47: [A]
Q48: [A]
Q49: [B]
Q50: [A]
Q51: [B]
Q52: [B]
Q53: [B]
Q54: [A]
Q55: [B]
Q56: [B]
Q57: [A]
Q58: [A]
Q59: [A]
Q60: [C]
Q61: [B]
Q62: [A]
Q63: [C]
Q64: [B]
Q65: [A]
Q66: [B]
Q67: [A]
Q68: [B]
Q69: [A]
Q70: [B]
Q71: [A]
Q72: [A]
Q73: [B]
Q74: [B]
Q75: [A]
Q76: [A]
Q77: [A]
Q78: [A]
Q79: [A]
Q80: [A]
Q81: [B]
Q82: [A]
Q83: [B]
Q84: [A]
Q85: [B]
Q86: [A]
Q87: [B]
Q88: [B]
Q89: [A]
Q90: [A]
Q91: [B]
Q92: [A]
Q93: [B]
Q94: [C]
Q95: [A]
Q96: [B]
Q97: [A]
Q98: [A]
Q99: [C]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

SECTION D: DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS

[illegible] sections (Introduction, Methods, Results, Discussion) [illegible] scientific discourse.

- Present Simple for overarching scientific claims, models, and general conclusions: "These data demonstrate that phosphorylation activates kinase activity".
- Past Simple when comparing current results with historical trials: "Our observed binding affinity was higher than the value reported by Jones (2020)".

>> **Solution to Question 4:** [illegible] modal substitutions.

- Moderate Probability: "If the primer anneals specifically, the amplicon should form."
- Cautious Epistemic Possibility: "If the primer anneals specifically, the amplicon may / might form."

[illegible]

Standard conditionals match temporal frameworks (Past-to-Past, Present-to-Future). Mixed conditionals bridge two distinct time frameworks, most commonly pairing a past counterfactual condition with an ongoing present result.

2. The Type A Mixed Conditional (Past Condition -> Present Result):
- Formula: If + [Past Perfect], ... [Would + Base Verb (Present)].
- Diagnostic Example in Biotechnology:
- Context: The lab failed to preserve an aliquot of a unique mutant strain in liquid nitrogen last year. Today, that strain is extinct.
- Mixed Formulation: "If the researchers had cryopreserved the hybridoma line in 2023, the monoclonal antibody would be available for our current clinical trials."
- Analysis: The condition is in the past ('had cryopreserved'), but the hypothetical consequence operates in the present ('would be available').

3. The Type B Mixed Conditional (Permanent Present Quality -> Past Consequence):
- Formula: If + [Past Simple / Subjunctive Were], ... [Would have + Past Participle].
- Example: "If the bacteriophage were not so lethal to Salmonella, the treatment would have failed during the outbreak."

[illegible]

- Simple Past + Explicit Chronological Anchor: When a sentence specifies an author, publication year, or historical date, the Simple Past is grammatically mandatory:
- Correct: "Sanger et al. developed dideoxy chain-termination sequencing in 1977."
- Incorrect: "*Sanger et al. have developed dideoxy sequencing in 1977."
- Present Perfect + General Unanchored Consensus: When referencing ongoing discoveries or collective knowledge without anchoring them to a discrete moment:
- Correct: "Numerous laboratories have successfully adapted Sanger sequencing for high-throughput diagnostics."
- Simple Present for Enduring Theories:
- Correct: "The central dogma of molecular biology states that genetic information flows from DNA to RNA to protein."

>> Solution to Question 10: [illegible] laboratory procedures.

- Grammatical: "The buffer solution consisted of Tris-acetate."

3. The Rare Legitimate Exception (Temporary Dynamic Process):
Some stative verbs shift to dynamic meaning when indicating temporary, active changes:
- "The optical density is increasing rapidly" (Dynamic change of state).

[illegible] biology papers.

networks." (Circulates remains present simple because circulation is a permanent anatomical fact).
- Modern Example: "Mendel proved that alleles segregate independently during gamete formation." (Segregate remains present simple).

[illegible]

[illegible]

- "Should the biohazard autoclave pressure exceed 2.2 bar, immediately engage the secondary emergency vent valve."
- "Should thermal runaway threaten the culture chamber, inject chilled liquid nitrogen through port B."
- "Should skin contact with acrylamide take place, flush the affected epidermis under copious running water for twenty minutes."

>> [illegible] 'as soon as').

tenses in both clauses must maintain logical temporal harmony:

2. Past Reference Frame:
- Antecedent Action (Past Perfect) + Subsequent Action (Past Simple):
- "After the researcher had calibrated the pH electrode, she initiated the enzymatic titration."
- "The fermentation proceeded smoothly until the dissolved oxygen level dropped below critical threshold."
- Note: When 'before' or 'after' establishes unambiguous temporal priority, both verbs may stand in Past Simple:
- "After the researcher calibrated the pH electrode, she initiated the titration."

3. Future Reference Frame:
- Independent Clause (Future Simple) + Dependent Time Clause (Present Simple):
- Never use future tense in the time clause:
- Ungrammatical: "We will harvest the biomass as soon as the optical density will peak."
- Grammatical: "We will harvest the biomass as soon as the optical density peaks."

[illegible]

3. Contextual Contrasts in Experimental Validation:
- When describing the historic creation of the model, use Past Simple:
- "Leonor Michaelis and Maud Menten derived this equation in 1913."
- When describing how the model behaves mathematically, use Present Simple:
- "The resulting curve asymptotically approaches Vmax."

[illegible]

1. Structural Anatomy:
- Formula: Subject + will be + [Verb + -ing].
- Meaning: Indicates an action that will be in active progress at a specific designated point or across an uninterrupted window in future time.

2. Application in Industrial Bioprocess Automation:
In automated pilot plants and biopharmaceutical manufacturing schedules, the Future Continuous communicates scheduled unattended automated cycles:
- "At midnight tonight, the automated bioreactor will be performing the primary feed cycle."
- "Throughout the weekend, the robotic liquid handler will be dispensing reagents into 384-well assay plates."
- "During the third week of the clinical phase, the bio-analytical unit will be profiling pharmacokinetic metabolites."

3. Strategic Advantage:
Emphasizes seamless operational momentum and professional project management in technical Gantt-chart documentation.

[illegible]

- "If a new batch of competence-certified E. coli arrives, the technician immediately tests transformation efficiency with pUC19 control plasmid."
- Past Habitual:
- "During early pilot runs, whenever foaming occurred in the fermenter, technicians added silicone antifoam manually."

3. Linguistic Distinction from Zero Conditional:
While zero conditionals express immutable physical laws of nature, habitual conditionals express regular operational protocols and behavioral conventions.

[illegible] in scientific peer reviews.

- Past Tense + Tentative Reporting Verb (Maximum Skepticism / Distance):
- "The previous authors alleged that the peptide cured leukemia in murine models."
- Stance: Strongly implies skepticism or lack of independent replication.
